



SAINT IGNATIUS' COLLEGE

HSC Trial

2001

MATHEMATICS

Extension 1

2:00 – 4:05pm
Wednesday 5th September 2001

Directions to Students

- Reading Time : 5 minutes
- Time Allowed : 2 hours
- Attempt ALL questions.
- Board approved calculators may be used.
- A standard integral table is provided
- Answer each question in a separate writing booklet and clearly label your name and teacher's name.

Total Marks 84

Attempt Questions 1 – 7

All questions are of equal value

Students are reminded that this is a trial examination only and cannot in any way guarantee the content or the format of the 2001 Mathematics Extension 1 Higher School Certificate examination

Total marks (84)

Attempt Questions 1 - 7

All questions are of equal value

Marks

Answer each question in a SEPARATE writing booklet.

QUESTION 1 (12 marks) Use a SEPARATE Writing Booklet.

- (a) When $x^3 - 3x^2 - 4x + k$ is divided by $(x + 2)$, the remainder is 3. Find the value of k . 2
- (b) The interval PQ has end points $P(5, -6)$ and $Q(-7, 10)$. Find the coordinates of the point R which divides PQ internally in the ratio 5 : 3. 2
- (c) Evaluate $\int_1^2 \frac{dx}{\sqrt{4-x^2}}$. 2
- (d) Solve $\frac{3}{2-x} > 1$. 3
- (e) If α, β, γ are the roots of the equation $x^3 - 5x^2 + 3x - 2 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. 3

Marks

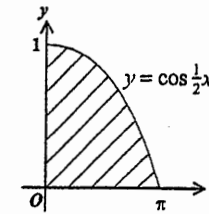
QUESTION 2 (12 marks) Use a SEPARATE Writing Booklet.

- (a) Using the substitution $u = x + 3$, find $\int_{-3}^{-2} x(x+3)^4 dx$. 3
- (b) Consider the function $f(x) = \sin^{-1}(x-1)$.
- (i) What is the domain of $y = f(x)$? 1
- (ii) Sketch the graph of $y = f(x)$. 1
- (c) Mr and Mrs Jones belong to a bush-walking club, which has a total of 20 members. A committee of 4 is chosen at random to plan the next bush-walk.
- What is the probability that:
- (i) both Mr and Mrs Jones will be on the committee? 2
- (ii) neither of Mr and Mrs Jones will be on the committee? 2
- (d) (i) Express $\tan 2\theta$ in terms of $\tan \theta$. 1
- (ii) By letting $\theta = \tan^{-1} 2$, prove that $2 \tan^{-1} 2 = \tan^{-1} \left(-\frac{4}{3} \right)$ 2

QUESTION 3 (12 marks) Use a SEPARATE Writing Booklet.

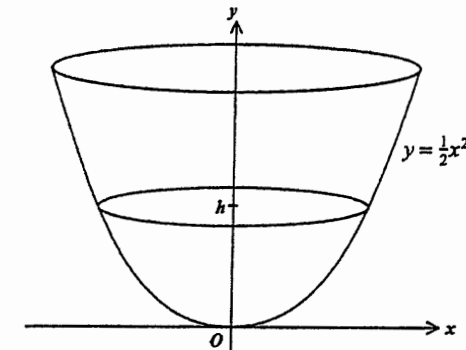
Marks

(a)



The diagram shows the graph of $y = \cos \frac{1}{2}x$, for $0 \leq x \leq \pi$.
The shaded area is rotated about the x -axis.
Find the volume of the solid formed.

(b)



A large industrial container is in the shape of a paraboloid, which is formed by rotating the parabola $y = \frac{1}{2}x^2$ around the y -axis.

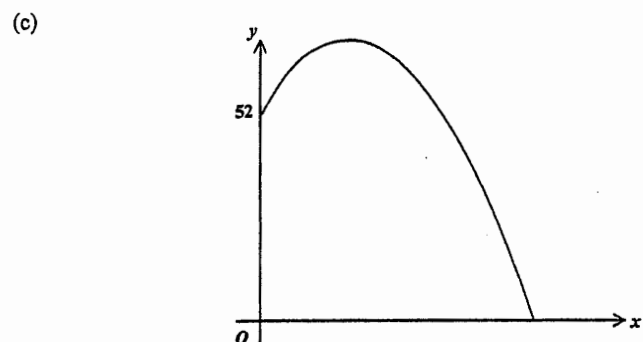
Liquid is poured into the container at a rate of 2 m^3 per minute.

- (i) Prove that the volume V of liquid in the container when the depth of liquid is h , is given by $V = \pi h^2$. 1
- (ii) At what rate is the height of the liquid rising when the depth is 1.5 m? 3
- (iii) If the container is 3 m high, how long will it take to fill the container? 1
- (c) Prove, using mathematical induction, that $5^n + 11$ is divisible by 4, where n is a positive integer. 4

QUESTION 4 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) Find the coefficient of x^2 in the expansion of $(3 + 2x)(2 + x)^6$. 3
- (b) A squad of 18 boys is selected for rugby training, from which a team of 15 players is to be chosen for the Saturday game.
The probability that a player will be injured at training and not available for Saturday is 0.15.
- (i) Find the probability that 3 players will be unavailable for the Saturday game. (Answer to 3 decimal places) 2
- (ii) Write the numerical expression for the probability that the team will not be able to field a team of 15 fit players on Saturday. 2
Do not simplify the answer.



A ball is projected from the top of a 52 metre high tower.
Its position t seconds after it is thrown, is given by the equations

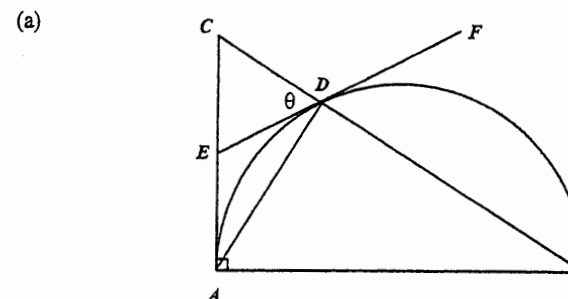
$$x = 12t, \quad y = 52 + 16t - 5t^2$$

where the origin O is on the ground vertically below the point of projection.

- (i) Find the greatest height reached above ground level. 2
- (ii) For what length of time is the ball in flight? 2
- (iii) How far from O does the ball land? 1

QUESTION 5 (12 marks) Use a SEPARATE Writing Booklet.

Marks



AB is the diameter of a semi-circle. $\triangle ABC$ is right-angled at A , and BC cuts the semi-circle at D . EF is a tangent to the semi-circle at D . AD is joined. $\angle CDE = \theta$.

COPY OR TRACE THE DIAGRAM ONTO YOUR WRITING PAGE.

- (i) Why is $\angle ADB = 90^\circ$? 1
- (ii) Why is $\angle ADE = \angle ABD$? 1
- (iii) Name two angles equal to $\angle CDE$. 1
- (iv) Prove $\triangle ADE$ is isosceles. 2
- (v) Prove that E is the midpoint of AC . 2
- (b) Consider the function $f(x) = (x - 2)^2 - 3$ for $x \leq 2$.
- (i) Sketch the function $y = f(x)$. 2
- (ii) Explain why $f(x)$ has an inverse function. 1
- (ii) Find the inverse function $y = f^{-1}(x)$. 2

27

QUESTION 6 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) The velocity of a particle moving in a straight line at position x is given by:
 $v = 2e^{-x}$.

Initially the particle is at the origin.

- (i) Show that the acceleration at position x is given by $a = -4e^{-2x}$.
 (ii) What is the initial acceleration?
 (iii) The position of the particle at time t is given by $x = \log_e f(t)$.
 Find the function $f(t)$.

2

1

2

- (b) Consider the binomial expansion, where n is an even number:

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n.$$

- (i) Prove that $\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots + \binom{n}{n-1} = 2^{n-1}$.

3

- (ii) Prove that $\sum_{r=1}^n r \binom{n}{r} = n \times 2^{n-1}$.

2

- (iii) Find an expression for $\sum_{r=0}^n (r+1) \binom{n}{r}$.

2

QUESTION 7 (12 marks) Use a SEPARATE Writing Booklet.

Marks

- (a) (i) Express $2 \sin t - 5 \cos t$ in the form $A \sin(t - \alpha)$, where α is in radians, $A > 0$.

2

- (ii) What is the amplitude of the function $f(t) = 2 \sin t - 5 \cos t$?

1

- (b) Show that the derivative of $8t \tan^{-1} 2t - 2 \log_e(1 + 4t^2)$ is $8 \tan^{-1} 2t$.

2

- (c) In the Olympic 100 metres running event, the speed v metres per second of a runner t seconds after the start is given by:

$$v = 8 \tan^{-1} 2t.$$

- (i) Using the result of part (b), explain why the time taken, T seconds, to complete the 100 metres is given by the equation

2

$$8T \tan^{-1} 2T - 2 \log_e(1 + 4T^2) - 100 = 0.$$

- (ii) Show that a root of this equation lies between $T = 9$ and $T = 10$.

2

- (iii) Using $T = 9$ as a first approximation, use Newton's method to find a better approximation, to one decimal place.

2

- (iv) Using this value of T , what is the runner's speed at the end of the 100m race?

1

End of paper



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3 Unit - Question 1.

$$(a) \quad P(x) = x^3 - 3x^2 - 4x + k$$

$$P(-2) = -8 - 12 + 8 + k$$

$$P(-2) = 3 \quad -12 + k = 3$$

$$k = 15$$

[2]

$$(b) \quad \begin{matrix} x_1 & y_1 & x_2 & y_2 & m_1 & m_2 \\ P(5, -6) & Q(-7, 10) & 5 & 3 \end{matrix}$$

$$R \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) = \left(\frac{5 \times (-7) + 3 \times 5}{5 + 3}, \frac{5 \times 10 + 3 \times (-6)}{5 + 3} \right)$$

$$= \left(-2\frac{1}{2}, 4 \right)$$

[2]

$$(c) \quad \int_1^2 \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_1^2$$

$$= \sin^{-1} 1 - \sin^{-1} \frac{1}{2}$$

$$= \frac{\pi}{2} - \frac{\pi}{6}$$

$$= \frac{\pi}{3}$$

[2]

$$(d) \quad \frac{3}{2-x} > 1$$

$$x(2-x)^2: \quad 3(2-x) > (2-x)^2$$

$$6 - 3x > 4 - 4x + x^2$$

$$x^2 - x - 2 < 0$$

$$(x-2)(x+1) < 0$$

$$-1 < x < 2$$



[3]

$$(e) \quad x^3 - 5x^2 + 3x - 2 = 0$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= -\frac{3}{2}$$

[3]

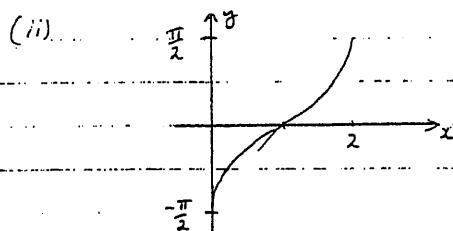
3 Unit - Question 2

$$\begin{aligned}
 (a) \int_{-3}^{-2} x(x+3)^4 dx &= \int_0^1 (u-3) u^4 du & u = x+3 \\
 &= \int_0^1 (u^5 - 3u^4) du & du = dx \\
 &= \left[\frac{1}{6} u^6 - \frac{3}{5} u^5 \right]_0^1 & \text{When } x = -3, u = 0 \\
 &= \left(\frac{1}{6} - \frac{3}{5} \right) - (0 - 0) & \text{When } x = -2, u = 1 \\
 &= -\frac{13}{30} & \boxed{3}
 \end{aligned}$$

$$(b) f(x) = \sin^{-1}(x-1)$$

$$(i) -1 \leq x-1 \leq 1 \quad \therefore 0 \leq x \leq 2$$

Domain is $0 \leq x \leq 2$



$$(c) (i) \text{ Prob} = \frac{\binom{18}{2}}{\binom{20}{4}} = \frac{153}{4845} = \frac{3}{95} \quad (\text{or } 0.032, 3dp) \quad \boxed{2}$$

$$(ii) \text{ Prob} = \frac{\binom{16}{4}}{\binom{20}{4}} = \frac{1820}{4845} = \frac{364}{969} \quad (\text{or } 0.376, 3dp) \quad \boxed{2}$$

$$(d) (i) \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad \boxed{1}$$

$$(ii) \text{ Let } \theta = \tan^{-1} 2, \therefore \tan \theta = 2$$

$$\tan 2\theta = \frac{2 \times 2}{1 - 2^2}$$

$$= -\frac{4}{3}$$

$$2\theta = \tan^{-1}\left(-\frac{4}{3}\right)$$

$$\therefore 2 \tan^{-1} 2 = \tan^{-1}\left(-\frac{4}{3}\right) \quad \boxed{2}$$

3 Unit - Question 3

Question 3

$$\begin{aligned}
 (a) V &= \pi \int_0^{\pi} y^2 dx \\
 &= \pi \int_0^{\pi} \cos^2 \frac{x}{2} dx \\
 &= \pi \int_0^{\pi} \frac{1}{2} (1 + \cos x) dx \\
 &= \frac{\pi}{2} [x + \sin x]_0^{\pi} \\
 &= \frac{\pi}{2} [(\pi + 0) - (0 + 0)] \\
 \text{Volume} &= \frac{\pi^2}{2} \text{ unit}^3. \quad \boxed{3}
 \end{aligned}$$

$$\begin{aligned}
 (b) (i) V &= \pi \int_0^h x^2 dy \\
 &= \pi \int_0^h 2y dy \\
 &= \pi [y^2]_0^h \\
 &= \pi h^2 \quad \boxed{1}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \frac{dV}{dt} &= \frac{dV}{dh} \times \frac{dh}{dt} \\
 \frac{dV}{dt} &= 2\pi h \times \frac{dh}{dt}
 \end{aligned}$$

$$\text{when } \frac{dV}{dt} = 2, h = 1.5$$

$$2 = 2 \times \pi \times (1.5) \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{2}{3\pi}$$

$$\text{Liquid is rising at } \frac{2}{3\pi} \text{ m/min}$$

$$\text{or } 0.212 \text{ m/min (3dp)} \quad \boxed{3}$$

$$(iii) \text{ When } h = 3, V = 9\pi$$

$$\text{Time taken} = \frac{9\pi}{2} \text{ minutes}$$

$$\text{or } 14.14 \text{ min (2dp)} \quad \boxed{1}$$

$$(c) \text{ Prove } 5^n + 11 \text{ is divisible by 4}$$

$$\text{When } n = 1, 5^n + 11 = 5 + 11$$

$$= 16$$

$\therefore$ It is true when $n = 1$.

Assume it is true for $n = k$

i.e. assume $5^k + 11 = 4I$ where I is integer

When $n = k+1$,

$$5^{k+1} + 11 = 5^{k+1} + 11$$

$$= 5^k \times 5 + 11$$

$$= 5(4I - 11) + 11 \text{ by assumption}$$

$$= 20I - 44$$

$$= 4(5I - 11)$$

which is divisible by 4

If it is true for $n = k$,

then it is true for $n = k+1$.

Since it is true for $n = 1$,

then it is true for $n = 2$

$\therefore$ it is true for $n = 3, \dots$

i.e. it is true for all positive

integers n . $\boxed{4}$

3 Unit - Question 4

Question 4

$$(a) (3+2x)(2+x)^6$$

$$= (3+2x) \left[\binom{6}{0} 2^6 + \binom{6}{1} 2^5 x + \binom{6}{2} 2^4 x^2 + \dots \right]$$

$$\text{Term in } x^2 = 3 \times \binom{6}{2} 2^4 x^2$$

$$+ 2x \times \binom{6}{1} 2^5 x$$

$$= 3 \times 15 \times 16 x^2 + 2 \times 6 \times 32 x^2$$

$$= (720 + 384) x^2$$

$$\text{Coefficient of } x^2 = 1104. \quad [3]$$

$$(b)(i) P = \binom{18}{3} (0.15)^3 (0.85)^{15}$$

$$= 0.241 \quad (3 \text{ d.p.}) \quad [2]$$

(ii) Probability of not fielding a full team

$$= P(4 \text{ or more injured})$$

$$= 1 - P(0 \text{ or } 1 \text{ or } 2 \text{ or } 3 \text{ injured})$$

$$= 1 - \left[\binom{18}{0} (0.85)^{18} + \binom{18}{1} (0.15) (0.85)^{17} \right. \\ \left. + \binom{18}{2} (0.15)^2 (0.85)^{16} + \binom{18}{3} (0.15)^3 (0.85)^{15} \right]$$

[2]

$$(c) \quad x = 12t, \quad y = 52 + 16t - 5t^2$$

$$\dot{x} = 12, \quad \dot{y} = 16 - 10t$$

(i) greatest height when $\dot{y} = 0$

$$16 - 10t = 0$$

$$t = 1.6$$

$$y = 52 + 16 \times 1.6 - 5 \times 1.6^2$$

$$= 64.8$$

Greatest height = 64.8 metres

[2]

(ii) Hit ground when $y = 0$

$$52 + 16t - 5t^2 = 0$$

$$5t^2 - 16t - 52 = 0$$

$$t = \frac{16 \pm \sqrt{256 - 4 \times 5 \times (-52)}}{10}$$

$$= \frac{-16 \pm 36}{10}$$

$$= 2 \text{ or } -5.2$$

Ball is in flight for 2 seconds [2]

(iii) when $t = 2, x = 12 \times 2$

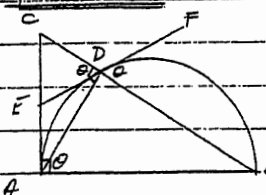
$$= 24$$

Ball lands 24 m. from O. [1]

3 Unit - Question 5

Question 5

(a)



(i) $\angle ADB = 90^\circ$ because it is an angle in the semi-circle [1]

(ii) $\angle ADE = \angle ABD$ because the angle between a tangent and a chord drawn to the point of contact is equal to an angle in the alternate segment. OR the alternate segment theorem. [1]

(iii) $\angle CDE = \angle FDB$ (vert. opp.)
 $= \angle DAB$ (alt. seg. thm) [1]

(iv) $\angle EDA = 90^\circ - \angle C$ from diagram
 $\angle EAD = 90^\circ - \angle C$ from diagram
 $\therefore AE = ED$ (sides opposite equal angles)

$\therefore \triangle ADE$ is isosceles. [2]

(v) Since $\angle EAD = 90^\circ - \angle C$
 then $\angle ACD = \angle C$ (angle sum of $\triangle ADC$)
 $\therefore \angle EDC = \angle ECD$
 $\therefore CE = ED$

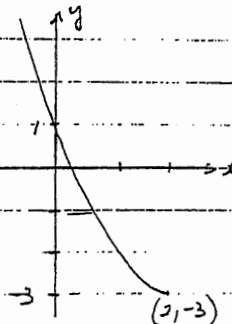
But $AE = ED$ in (iv)

$\therefore AE = CE$

$\therefore E$ is midpoint of AC . [2]

(b) $f(x) = (x-2)^2 - 3$ for $x \leq 2$

(i)



[2]

(ii) $f(x)$ has an inverse because it is a 1:1 relation

i.e. for each value of x there is one value of y and for each value of y there is one value of x .

OR it satisfies the horizontal line test. [1]

(iii) Inverse is

$$x = (y-2)^2 - 3 \text{ for } y \leq 2$$

$$x+3 = (y-2)^2$$

$$y-2 = \pm \sqrt{x+3}$$

$$y = 2 \pm \sqrt{x+3}$$

But $y \leq 2$

$$\therefore y = 2 - \sqrt{x+3}$$

$$\text{or } f^{-1}(x) = 2 - \sqrt{x+3}. \quad [2]$$

3 Unit - Question 6.

Question 6.

(a.) $v = 2e^{-x}$

(i) $a = \frac{dv}{dx} \left(\frac{1}{2} v^2 \right)$
 $= \frac{d}{dx} \left(\frac{1}{2} \times 4e^{-2x} \right)$
 $= 2 \times (-2) e^{-2x}$
 $= -4e^{-2x}$ [2]

(ii) When $t=0$, $x=0$

$\therefore a = -4e^0$
 $= -4$

Initial acceleration is -4 units.

[1]

(iii) $\frac{dx}{dt} = 2e^{-x}$
 $= \frac{2}{e^x}$

$\frac{dx}{dt} = \frac{e^x}{2}$

$t = \frac{1}{2} e^x + C$

When $t=0$, $x=0$

$\therefore 0 = \frac{1}{2} e^0 + C$
 $C = -\frac{1}{2}$

$\therefore t = \frac{1}{2} e^x - \frac{1}{2}$

$2t = e^x - 1$

$e^x = 2t + 1$

$x = \ln(2t + 1)$

$\therefore f(t) = 2t + 1$ [2]

(b) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \dots + \binom{n}{n}x^n$
 $(n \text{ is even})$

(i) Let $x = -1$.

$0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots - \binom{n}{n-1} + \binom{n}{n}$
 $\therefore \binom{n}{1} + \binom{n}{3} + \dots + \binom{n}{n-1} = \binom{n}{0} + \binom{n}{2} + \dots + \binom{n}{n}$

Let $x = 1$:

$2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}$
 $\therefore \binom{n}{1} + \binom{n}{3} + \dots + \binom{n}{n-1} = \frac{2^n}{2}$
 $= 2^{n-1}$ [3]

(ii) Differentiate:

$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \dots$
 $\dots + n\binom{n}{n}x^{n-1}$

Let $x = 1$:

$n \times 2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + \dots + n\binom{n}{n}$
 $\therefore \sum_{r=1}^n r\binom{n}{r} = n \times 2^{n-1}$ [2]

(iii) $x(1+x)^n = \binom{n}{0}x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \dots$
 $\dots + \binom{n}{n}x^{n+1}$

Differentiate:

$(1+x)^n + x n(1+x)^{n-1}$
 $= \binom{n}{0} + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + \dots$
 $\dots + (n+1)\binom{n}{n}x^n$

Let $x = 1$:

$2^n + n \times 2^{n-1} = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots$
 $\dots + (n+1)\binom{n}{n}$

$\therefore \sum_{r=0}^n (r+1)\binom{n}{r} = 2^n(2+n)$ [2]

3 Unit - Question 7.

Question 7.

(a) (i) $2 \sin t - 5 \cos t = A \sin(t-\alpha)$

$= A \sin t \cos \alpha - A \cos t \sin \alpha$

Equating coefficients of $\sin t, \cos t$:

$A \cos \alpha = 2$ (1)

$A \sin \alpha = 5$ (2)

$A^2 \cos^2 \alpha + A^2 \sin^2 \alpha = 2^2 + 5^2$

$A^2 (\cos^2 \alpha + \sin^2 \alpha) = 29$

$A = \sqrt{29}$ ($A > 0$)

From (1) (2), since $A > 0$, α is acute.

(2) $\div$ (1) $\tan \alpha = \frac{5}{2}$

$\alpha = 1.107$

$\therefore 2 \sin t - 5 \cos t = \sqrt{29} \sin(t - 1.107)$

[2]

(ii) Amplitude of $f(t)$ is $\sqrt{29}$.

[1]

(b) (i) $\frac{d}{dt} [8t \tan^{-1} 2t - 2 \log_e(1+4t^2)]$

$= \tan^{-1} 2t \times 8 + 8t \times \frac{2}{1+4t^2} - 2 \times \frac{8t}{1+4t^2}$

$= 8 \tan^{-1} 2t$

[2]

(c) (i) $v = 8 \tan^{-1} 2t$

$u = \int 8 \tan^{-1} 2t \, dt$

$= 8t \tan^{-1} 2t - 2 \log(1+4t^2) + C$

When $t=0$, $x=0$

$0 = 0 - 2 \log 1 + C$

$\therefore C = 0$

$x = 8t \tan^{-1} 2t - 2 \log(1+4t^2)$

When $x=100$, $t=T$

$\therefore 100 = 8T \tan^{-1} 2T - 2 \log(1+4T^2)$

i.e. $8T \tan^{-1} 2T - 2 \log(1+4T^2) - 100 = 0$

[2]

(ii)

$f(9) = 8 \times 9 \tan^{-1} 18 - 2 \log_e(1+4 \times 9^2) - 100$
 $= -2.466$

$f(10) = 8 \times 10 \tan^{-1} 20 - 2 \log_e(1+4 \times 10^2) - 100$
 $= 9.679$

Since $f(9)$ and $f(10)$ have opposite signs, a root lies between 9, 10.

[2]

(iii)

$T = 9 - \frac{f(9)}{f'(9)}$

$= 9 - \frac{(-2.466)}{8 \times \tan^{-1} 18}$

$= 9 - \frac{(-2.466)}{12.122}$

$= 9.2$ (1 d.p.)

[2]

(iv) When $T = 9.2$,

$v = 8 \tan^{-1}(2 \times 9.2)$

$= 12.13$

Runner's speed is 12.13 m/s. [1]

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